

ZenoTalent

AI Clinical Workflow Copilot

Product Development Instructions (Client Demo Ready)

Product Objective

Develop an **AI-powered Clinical Workflow Copilot** that assists clinicians by automatically capturing consultations, generating structured clinical documentation, and supporting clinical workflow tasks.

The system should demonstrate how AI can **reduce administrative workload for doctors by automating documentation and clinical workflows**.

The platform should be designed as a **SaaS healthcare solution suitable for UK clinics and NHS environments**.

Key Product Capabilities

1. Consultation Capture

The system must support:

- Voice-based consultation recordings
- Text-based consultation input
- Audio upload for processing

Goal: Allow doctors to easily capture patient conversations.

Technology:

- Whisper Speech-to-Text

2. Medical Transcription

Convert doctor–patient conversations into **accurate clinical transcripts**.

Requirements:

- High-quality transcription
- Medical terminology support
- Speaker separation (optional)

Technology:

- Whisper
- Python processing pipeline

3. Clinical Information Extraction

The system must extract key medical data from transcripts:

- Patient information
- Symptoms
- Diagnoses
- Medications
- Allergies
- Vital signs
- Medical history

Technology:

- spaCy
- SciSpacy
- Custom Medical NER models

4. AI Clinical Documentation Generation

The platform should automatically generate structured documentation such as:

- SOAP Notes
- Consultation Summary
- Patient Visit Notes

Technology:

- OpenAI / LLM APIs
- Prompt engineering for medical note formatting

5. Clinical Workflow Copilot

Beyond documentation, the AI Copilot should assist doctors with workflow tasks:

Automatically generate:

- Referral letters
- Follow-up instructions
- Prescription summaries
- Discharge notes
- Clinical coding suggestions (SNOMED / ICD)

This feature differentiates the product from basic **AI scribe tools**.

6. Doctor Dashboard

Develop a web-based dashboard where clinicians can:

- Upload consultations
- View transcripts
- Review AI-generated documentation
- Edit reports
- Export clinical notes

Technology:

- Streamlit (for MVP demo)
- or
- React (for scalable production)

7. Healthcare Insights Dashboard

Provide simple analytics for clinics:

- Symptom trends
- Medication frequency
- Consultation statistics
- Patient visit patterns

Technology:

- Pandas
- Plotly / Streamlit charts

Technology Architecture

Backend

Python

FastAPI

Speech Processing

Whisper

Natural Language Processing

spaCy / SciSpacy

AI Generation

OpenAI / LLM APIs

Frontend

Streamlit (Demo version)

Data Processing

Pandas

Database

SQLite / PostgreSQL

System Workflow

Step 1

Doctor uploads consultation audio

Step 2

Speech-to-text engine converts audio to transcript

Step 3

NLP engine extracts medical entities

Step 4

AI model generates structured clinical documentation

Step 5

Clinical Copilot generates additional workflow outputs:

- referral letter
- follow-up instructions
- discharge summary

Step 6

Doctor reviews results on dashboard

Step 7

Reports can be exported or saved

Development Responsibilities

Backend Developer

- Build FastAPI services
- Manage file uploads and processing
- Integrate NLP and AI modules

AI Engineer

- Implement transcription pipeline
- Integrate LLM note generation

NLP Engineer

- Build medical entity extraction models
- Improve entity recognition accuracy

Frontend Developer

- Build doctor dashboard
- Display transcripts and AI outputs
- Enable report editing

Data Analyst

- Build healthcare analytics dashboards

Client Demo Flow

During the demo:

- 1 Upload doctor–patient conversation
- 2 AI converts audio to transcript
- 3 NLP extracts symptoms, medications, patient details
- 4 AI generates structured clinical notes
- 5 Copilot generates:
 - referral letter
 - follow-up instructions
 - clinical summary
- 6 Doctor reviews everything in the dashboard

This demonstrates **AI-powered clinical workflow automation**.

Expected Outcome

A working prototype demonstrating:

- AI consultation transcription
- automatic clinical documentation
- AI workflow assistance for clinicians
- interactive doctor dashboard
- healthcare analytics

The system should clearly show how **AI reduces documentation workload for healthcare professionals**.

Core AI Copilot Features (For Client Demo)

1. Consultation Notes Generator

The AI automatically converts doctor–patient conversations into **structured clinical notes**.

The system processes consultation transcripts and generates formats such as:

- SOAP Notes (Subjective, Objective, Assessment, Plan)
- Consultation Summary
- Patient Visit Documentation

This eliminates the need for doctors to manually write consultation notes.

Example Output

Patient Complaint: Dizziness for 3 days

Symptoms: Vertigo, nausea

Assessment: Possible vestibular disorder

Plan: Refer to Neurologist, prescribe medication

2. Smart Referral Letter Generator

The AI Copilot automatically creates **specialist referral letters** based on consultation data.

If the patient requires specialist care, the system generates a referral document for the appropriate department.

Examples:

Patient Issue	Specialist Referral
Dizziness / Neurological symptoms	Neurologist

Knee pain / joint injury	Orthopedic Specialist
Dental pain	Dentist
Heart symptoms	Cardiologist
Eye problems	Ophthalmologist

Example scenario:

Patient reports **chronic neurological symptoms treated previously in India**.

The doctor decides to refer the patient to an NHS neurologist.

The AI generates a **referral letter automatically**.

Example Output:

Dear Neurology Department,

The patient presents with persistent dizziness and has a prior neurological treatment history. Please evaluate for further diagnosis and management.

This reduces **manual administrative work for clinicians**.

3. SNOMED Coding Automation

Doctors often dislike manually selecting clinical codes for documentation and NHS reporting.

The AI Copilot automatically **suggests relevant SNOMED codes** based on diagnosis.

Example:

Diagnosis: Diabetes Mellitus

Suggested SNOMED Code: **44054006**

Diagnosis: Knee Pain

Suggested SNOMED Code: **399153001**

Diagnosis: Migraine

Suggested SNOMED Code: **37796009**

This improves:

- NHS reporting
- medical record accuracy
- billing documentation

Doctors can simply **confirm the suggested code**.

4. Patient Summary Generator

The AI automatically generates a **clear patient summary** after each consultation.

This helps doctors quickly review patient history during future visits.

Example Summary:

Patient: John Smith

Complaint: Dizziness and balance issues

History: Previous neurological treatment in India

Assessment: Suspected vestibular disorder

Plan: Referral to Neurology Department

Benefits:

- Faster follow-up consultations
- improved patient record clarity
- reduced documentation time